S Ayushmaan Patro

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.37
Khallikote Higher Secondary School	Class XII (CHSE)	2023	85.66%
DAV Public School	Class X (CBSE)	2021	96.20%

SCHOLASTIC ACHIEVEMENTS

Received the **Academic Excellence Award** for securing the **highest SGPA in the Computational Engineering** branch in both **2023** and **2025.**

Currently holding **Department Rank 1** in Computational Engineering, batch of 2027.

Achieved an **SGPA** of **10.00** in the 5th Semester.

Co-developed and **maintained** the **IITH Computational Department Website.**

WORK EXPERIENCE

Morgan Stanley *(Application Developer Intern)*

- **Developed** a full-stack enterprise reporting platform using **Angular 21** and **Spring Boot**, enabling users to browse, filter, and execute **500+** parameterized reports with one-click CSV export.
- Implemented **RBAC** with **OIDC/PingFederate**, designing a proxy-based authentication flow that propagated trusted user identity to the Spring Security backend.
- **Containerized** the application with **Podman** and automated deployments through a **Jenkins CI/CD** pipeline, streamlining delivery and ensuring consistent releases.

Scale AI *(AI Trainer at Outlier.ai)*

- **Audited** and **optimized** AI-generated C++/Python code and mathematical reasoning, identifying logical errors, hallucina tions, and implementation issues to improve the quality of **LLM** training data.

PROJECTS

IITH Complaints Website (Group Project)

- **Contributed** to a complaint management platform supporting **400+** hostellers and over **50** facility-related complaints, reducing average resolution time by **40%**.
- Included Google **OAuth 2.0** authentication, supporting secure user logins with **99.9%** uptime.
- **Automated** email notifications to relevant authorities using **Google Apps Script**, reducing manual outreach by **90%** and improving response acknowledgments by **50%**.

Real-Time Chat Application

- **Built** a full-stack real-time messaging application using **React** and **Node.js**, organising the backend into **6+** modular layers for controllers, middleware, models, routes, external services, and utilities.
- **Developed** interactive chat functionality with **Socket**-based real-time communication, authentication workflows, message handling, and centralised client-side state management across **3** core user-facing pages.
- **Integrated** services for media handling, email workflows, and **API** security, while building a reusable React UI architecture with **15+** components for conversations, contacts, profiles, loading states, and messaging.

Dynamic Sudoku Validation using Multithreading

- Wrote a **C++** program to check Sudoku boards using multiple threads for faster results.

- **Optimised** thread synchronisation using atomic operations for correctness and early termination

Producer-Consumer Problem

- **Simulated** the Producer-Consumer problem in C++ using semaphores and custom locks.
- **Demonstrated** thread coordination and mutual exclusion using classic concurrency mechanisms.

GPT-2-Core (Micro-LLM Architecture)

- **Structured** an end-to-end **GPT-2** model from scratch in PyTorch, implementing causal multi-head self-attention, layer normalization, and residual connections.
- **Engineered** a data pipeline using tiktoken BPE tokenisation and custom PyTorch DataLoaders for sub-word tokenisation and dynamic tensor batching.
- **Executed** an inference pipeline with cross-entropy loss tracking, temperature scaling, and top-K sampling to study decoding strategies and text generation quality.

SKILLS

Programming Languages: C, C++, Python, Java, JavaScript, HTML, CSS

Frameworks & Libraries (Familiar): Angular, Spring Boot, React.js, Next.js, Node.js, Express.js, Tailwind CSS, PyTorch

Developer Tools (Familiar): Git, Docker, Podman, Jenkins, REST APIs, OAuth 2.0, OIDC, PingFederate, Google Apps Script

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures & Applications, Operating Systems, Foundations of Machine Learning.

Mathematics (All Introductory courses): Calculus, Linear Algebra, Differential Equations, Probability, Transform Techniques, Statistics, Complex Variables, Numerical Methods